Fiscal Year:	FY 2007	**Task Last Updated:**	FY 10/26/2007
PI Name:	Tang, Jay X Ph.D.		
Project Title:	Microscopic Studies of Gravi-Sensitive Microtubule Assembly in Simulated Variable Gravity Conditions		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	06/01/2004	**End Date:**	03/31/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NCE to 3/31/2007, per A. Chu 6/23/06		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Valles, Jr., James (Brown University)		
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	Dynamic assembly of tubulin is an essential biological process of fundamental importance to cell morphology, division and growth. We are investigating the mechanism of pattern formation during microtubule assembly, with the goal of understanding why polymerized microtubules form striated patterns under normal gravity but not under microgravity. A quantitative microscopic picture of the evolution of the patterns is acquired in the first phase of the program. This detailed characterization is used to test physical models of the process. The second phase of the study is to test how small perturbations, such as altered gravity, affect the pattern formation process. We will employ a novel Variable Magnetic Force (VMF) apparatus, the capabilities of which are such that the forces that couple to a solution's density fluctuations can be tuned to vary from microgravity to hypergravity. The threshold for the gravitational response of pattern formation polymerizing microtubules will be investigated using this instrument. By examining how sensitive these patterns are to		

Task Description:	altered gravity, new insight into the effects of gravity in determining the biologically relevant structures will be obtained. The results of the program may also lead to effective countermeasures to the adverse effects of microgravity on tissue and organ development, as well as alterations in human physiology and fertility in long term space travels. Our first goal is to understand how polymerized microtubules form striated patterns under normal gravity but not under microgravity as has been reported by Tabony and co-workers (Papaseit et. al., PNAS, 2000, 97, 8364). We have reproduced most of the observations by Tabony et. al. under normal gravity conditions. The formation of the patterns is strongly influenced by a transient, convective flow, induced by either an initial temperature gradient or a static magnetic field. Similar patterns were reproduced by shearing polymerizing tubulin samples, suggesting interplay between flow alignment and the continuous growth of microtubules and microtubule bundles following the rapid initial assembly of tubulin. Thus the physical mechanism of tubulin pattern formation is most likely attributable to coupling between gravity and convection, which profoundly affects all levels of biological and physiological functions in space travel. The objective of our ongoing investigation is threefold: 1, to precisely define the role of gravity on pattern formation; 2, to decouple the effects of convection and gravity using the Variable Magnetic Force (VMF) apparatus; and 3, to design a strategy for simulated gravity or certain desirable effects of convection in space travel environment. Convection is one of the fundamental mechanisms for heat transport and thermal equilibration. It works in a container of fluid, as well as within the tissues and organs in our body. Since the warmer liquid tends to be lighter, thermal inhomogeneity efficiently dissipates on earth, facilitated by the coupling between convection and gravity. Our body is well adapted during the course of evolution on earth to utilize this physical mechanism, but the possible adverse effects due to lack of this mechanism in space travel are poorly recognized. Among the 45 risks listed in the NASA Bioastronautics Roadmap for human space exploration, our study is relevant to Risk 4. Renal stone formation, and Risk 40. Maintain thermal balance. Current countermeasures listed include enhanced fluid uptake and selection of thermal transporting fluids to address the two risks, respectively. A better strategy to facilitate flow and fluid transport both within the human body and in the space environment is crucial for space travel. Our study uniquely addresses this important issue using a protein matrix with essential biological functions including cell division and tissue development. The ground based research program we conduct may also be relevant to Risk 2-Impaired fractural healing, Risk 6-Diminished cardiac and vascular function, and a yet to be addressed risk issue, namely, reproduction in long term space exploration.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This ground based research program defines the physical and biochemical mechanisms of tubulin assembly, which plays important functional roles in cell division and tissue development. The relevant mechanism bears resemblance to and suggests potential side effects of certain human fertility treatment. The results of this study may suggest useful measures to alleviate the side effects. Additionally, this research program explores the effects of a high strength, high gradient magnetic field on tubulin assembly. This approach demonstrates a way of manipulating biological samples that is underutilized at the present.
Task Progress:	The project has been carried out as planned and has yielded insightful and valuable results over its grant period. The results have been published in Proceedings of National Academy of Science (PNAS, 103:10654, 2006) and in Physical Review Letters (PRL, 98:198103, 2007). Two graduate students under the NASA support, Mr Yongxing Guo and Ms Yifeng Liu, performed a number of experiments on Microtubule (MT) assembly and pattern formation under static magnetic fields as specified in Specific Aims 1 and 2. Meanwhile, the assembled patterns were subjected to detailed microscopic characterization and quantitative analysis. We now understand the mechanism of the pattern forming process, which can explain the origin of the periodic property of the patterns and predict the periodicity based on a mechanical instability model. Our results imply a novel mechanical mechanism for biological pattern formation and morphogenesis, thus unveiling the mystery that has long existed regarding striped patterns in MT solutions. This mechanism might account for some of the global aspects of biological morphogenesis and development. The PI and co-PI each attended a few scientific meetings within the grant period, presenting research findings funded by NASA. The PI delivered an invited talk to the 4th World Congress of Cellular and Molecular Biology, on the mechanism of tubulin assembly. This congress, held in October, 2005, Poitiers, France, featured 10 Nobel Laureates in attendance. The PI also gave invited seminars in the Physics Department, Lehigh University, and Courant Institute, New York University, and Marine Biological Laboratory, Woods Hole, MA, on the NASA sponsored research program. The co-PI delivered talks at two international meetings for the use of high magnetic fields to manipulate materials systems. At both, the Symposium on Behavior of Substances and Organisms under High Magnetic Fields, Physical Phenomena at High Magnetic Fields–V, Tallahassee, Florida, August 5-9, 2005 and the 2nd International Workshop on Materials Analysis and Processing in Magnetic Fields, March 19-22, 2006, CNRS, Grenoble, France, the co-PI delivered talks discussing the influence of magnetic fields on microtubules. Graduate student Mr. Yongxing Guo, who was supported by the NASA award, gave a contributing talk in the American Physical Society Annual Meeting in March 2006, on a mechanic model that explains the MT patterns. The second graduate student supported by the NASA project, Ms. Yifeng Liu, attended the Biophysical Society Annual Meeting in Feb, 2006, and presented a poster on the NASA funded work.
Bibliography Type:	Description: (Last Updated: 11/01/2007)
Articles in Peer-reviewed Journals	Guo Y, Liu Y, Tang JX, Valles JM Jr. "Polymerization force driven buckling of microtubule bundles determines the wavelength of patterns formed in tubulin solutions." Phys Rev Lett. 2007 May 11;98(19):198103. PMID: 17677665 , May-2007
Articles in Peer-reviewed Journals	Liu, Y, Guo, Y, Valles, JM, Tang, JX. "Microtubule bundling and nested buckling drive stripe formation in polymerizing tubulin solutions." Proc Natl Acad Sci U S A. 2006 Jul 11;103(28):10654-9. PMID: 16818889 , Jul-2006